

Student prefixes
+ teacher targets



Task gradients
 μ_i, Σ_i



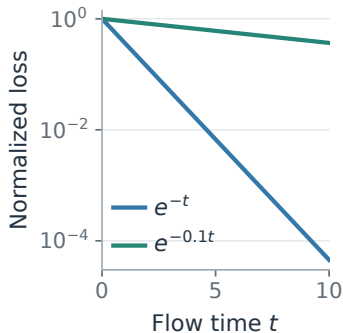
Fixed w, D : mean a_i
Counts m : variance v_i

Exact mean updates

— $G = 4$

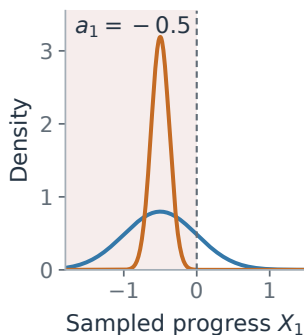
— $G = 64$

(a) Unequal rates



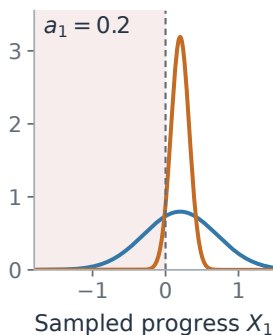
All $a_i > 0$; zero conflict
No sampling noise

(b) Harmful mean



$\Pr(X_1 \leq 0), G: 4 \rightarrow 64$
84.1% $\rightarrow$ 99.997%

(c) Useful mean



$\Pr(X_1 \leq 0), G: 4 \rightarrow 64$
34.5% $\rightarrow$ 5.5%

Analytical examples, not MOPD measurements. Shading: $X_1 \leq 0$.